BORA AYHAN

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Education

THE UNIVERSITY OF TEXAS AT AUSTIN **May 2028**
Bachelor of Science in Mechanical Engineering; Minor in Robotics
- Robotics Honors, Engineering Honors, ASME Member, AIAA Member

Skills

Python, C++, Java, Version control (Git), Linux, ROS2, Sensor Fusion, PID, CoDR/PDR/CDR, MS Office, Task Ownership

Experience

SOFTWARE INTERN **Jun. 2025 – Aug. 2025**
Autonomy, Robotics, and Intelligent Algorithms (ARIA) Laboratory Golden, CO
- Published open-source drivers, packages, Dockerized computer vision and SLAM pipelines using ROS2, and internal documentation for the Boston Dynamics *Spot*, Clearpath *Husky*, and the ARIA Lab Sensor Backpack.
- Collected over 50GB of LiDAR, camera, and odometry data (rosbags) for sensor fusion research by directing on-site experiment and demonstration of SLAM software at Edgar Experimental Mine, simulated on Gazebo.

SOFTWARE ENGINEER – GUIDANCE, NAVIGATION, AND CONTROL (GNC) **Sep. 2026 – present**
Texas Aerial Robotics, Raytheon Competition Team Austin, TX
- Configured NVIDIA Jetson (Linux) to interface with LiDAR, QGroundControl, and network telemetry router

SIMULATION/VALIDATION ENGINEER **Sep. 2026 – present**
Formula SAE (Longhorn Racing – Internal Combustion) Austin, TX
- Designed and implementing PostgreSQL team database with simulation workflow automation.
- Presented PDR on tuning quasi-static lap time simulator (QSS) deltas with empirical data for accuracy.

LEAD PROGRAMMER (VISION TRACKING) **Dec. 2023 – May 2026**
FIRST Robotics Team 4293 Louviers, CO
- Increased team's average points scored per match by 60% by developing sensor fusion targeting pipeline combining vision (ARUCO tags), IMU, and swerve drive encoders in a triple-redundant subsystem in Java.
- Owned pull request reviews to ensure reliability (GitHub), achieving 100% code uptime in competitions.

FOUNDER, PRESIDENT **Apr. 2024 – May 2026**
Arapahoe Rocketry Club Littleton, CO
- Led a team of 15 students in the design, production, and testing of a thrust-vector-controlled (TVC) rocket.
- Architected TVC system with PID/sensor fusion (Kalman-filtered IMU/barometer) around an FSM. (GitHub)
- Organized internal PDR and external industry CDR with CONOPS, state diagrams, and risk assessment.

DEPRESSURIZATION ENGINEER **Sep. 2026 – present**
Texas Guadaloop Austin, TX

RESEARCHER **Aug. 2026 – present**
Group for Artificial Intelligence (AI) in Materials Modeling and Analytics (GAMMA) Lab Austin, TX
- Researching AI-assisted particle characterization from SEM images using segmentation models with TACC.

Independent Projects

Trim, Auto Land, Odometry, and Navigation (TALON) Flight Computer **Jul. 2026 – Aug. 2026**
C++ flight software with autopilot functionality for Class 1 UAVs, in development. (GitHub)
- Implements Dubins path generation and L1 guidance (Park, Deyst, & How, 2004) for flight path execution.
- Architected minimally coupled flight software that isolates guidance and input logic from actuator subsystem.

Design and Assembly of a Dual Propeller Fixed Wing UAV **Jun. 2026 – Jul. 2026**
60-inch wingspan, <$45 per unit, positively longitudinally stable drone with 3 kg MTOW
- Integrated wing-embedded assembly (80 g) joining motor, ESC, battery, and tail boom.
- Electronics and radio embedded entirely within airfoil structure to reduce parasitic drag.
- 3D-printed wingbox and rib/spar structure to maintain Clark-Y airfoil at 5° dihedral for roll stability.

Awards

Lockheed Martin "CodeQuest" Programming Competition, Third Place (Python) **2023**
School of Mines Programming Competition, Second Place (Python) **2024**